DFlash vs. baseline — qwen3.5-9b on LongBench-E, hidden_states=selective, phase-instrumented sweep
4K / 8K / 16K / 32K / 64K · draft-only metrics stay in plot_summary_v2.py

Samples per context length (qwen3.5-9b) — 4K: n=32 (13 datasets) · 8K: n=32 (12 datasets) · 16K: n=32 (9 datasets) · 32K: n=32 (9 datasets, 31 composed) · 64K: n=32 (8 datasets, 32 composed)
Mean output tokens per request [DFlash / baseline] — 4K: 246 / 178 · 8K: 185 / 149 · 16K: 246 / 230 · 32K: 24 / 11 · 64K: 24 / 11 (long-context runs emit far fewer tokens; compare per-token metrics, not per-request ones)

Panels highlighted in green are new or re-measured relative to plot_vs_baseline.py (v1, same records directory, pre-instrumentation sweep) — ADDED = a quantity the v1 records did not carry, REVISED = same quantity, measured at a different instant

[1] ADDED summary.*.phase_memory, a new record field, for both configurations. The run is cut into named phases -- twelve for DFlash, the five the baseline actually runs -- and each records its own interval peak plus the component split probed at the occurrence that peaked.

[2] REVISED Target KV now has two numbers. max_target_cache_gb is _cache_bytes() called once at end of sample, which is what v1 plotted; peak_components_gb['target_kv_bytes'] is the same cache measured when the peak happened. They agree on Qwen3-8B and do not on Qwen3.5-9B, whose gated delta-rule layers hold a sequence-length prefill buffer that is released on the first decode forward -- so the end-of-sample reading misses up to 24 GB that was resident through the whole peak.

[3] ADDED The DFlash-minus-baseline peak delta, against selected_hidden at the peak. Both configurations load the drafter's weights and both run the same target prefill, so the delta is neither of those; it is the injected target hidden states, the one DFlash term alive while the target is still prefilling. The ratio is 1.000 on Qwen3-8B and 7/8 on Qwen3.5-9B, where one of the eight injected layers' residual streams is a tensor the target's own forward had allocated regardless -- the tap retains it rather than adding it.

[4] REVISED Device count differs from the full-mode records: 9B/32K 2->1, 9B/64K 3->2. Selective frees enough memory to fit on fewer cards. 9B/64K is still sharded and starred on the x axis, so its timings are pipeline-parallel. Memory stays comparable across a device change; timings do not -- going from sharded to single-GPU speeds the *baseline* up more than it speeds DFlash, so a speedup can read lower without DFlash having got slower.

[5] REVISED 9B/64K is sharded over more than one card (starred on the x axis). Per-stage draft timings are recorded with CUDA events on the drafter's stream, so a stage whose weights live on the other card -- lm_head, in practice -- is timed as the launch, not the work: it reads as ~0.05 ms against ~2.8 ms on a single card. Read the stage split of a starred column qualitatively only, and never subtract it from a single-GPU column.